FULL-STACK AI PLATFORM

ElevAI

AI-Powered Smart Interview & Career Preparation Portal

An end-to-end career intelligence and recruitment portal bridging job seekers and hiring teams through multi-dimensional ATS resume intelligence, AI mock interviews, and automated recruitment pipelines.

&›þ React 18 & Tailwind CSS

Ø=Þ€ Node.js & Express REST API

Ø=Ü Supabase PostgreSQL & Prisma

'(Google Gemini 2.0 Generative AI

"Your Career, Elevated."

Project Overview & Core Mission

A Unified Platform for Candidates & Hiring Teams

Ø=ÜÄ Resume Intelligence

Evaluates ATS compatibility (0–100 score), section structure, and rewires passive bullet points into quantifiable Google X-Y-Z achievements.

Ø<ß¯ Job-Specific Fit

Computes mathematical match scores (0–100%) against live job descriptions, highlighting keyword density, matching skills, and critical gaps.

Ø<ß™þ AI Mock Interview

Simulates dynamic Technical, HR, and Mixed interview sessions with automated feedback across technical depth, clarity, and ideal answers.

Ø>Ýí Skill Gap Roadmaps

Extracts existing competencies and builds structured role readiness assessments with prioritized learning action items.

Ø=ÜÝ Interactive Builder

Drag-and-drop section customization, ATS preview, and direct single-click compilation into the active user resume vault.

Ø<ßâ Recruiter Control

Full job lifecycle management (Draft -> Publish -> Close), candidate review dossiers, private notes, and real PostgreSQL analytics.

Problem Statement

Bridging the Critical Disconnect in Talent Acquisition

& þ Candidate / Student Challenges

• Generic Static Resumes: Candidates submit identical resumes across diverse jobs, failing ATS keyword filters.

• Ambiguous Skill Gaps: Job descriptions outline broad requirements; applicants do not know exact missing tools.

• Cost-Prohibitive Mock Interviews: Personalized technical coaching is expensive and inaccessible to most students.

• Disjointed Preparation: Switching between word processors, job boards, and tutorials causes fragmented career progress.

Ø<ßâ Recruiter & Placement Pain Points

• Manual Spreadsheet Filtering: Evaluating hundreds of unstructured resumes leads to severe review fatigue.

• Opaque Candidate Match: Recruiters lack immediate skill match snapshots and quantitative compatibility metrics.

• Disorganized Status Tracking: Candidate communications and stage updates get lost in fragmented email threads.

• No Real-Time Telemetry: Lack of integrated recruitment analytics to track 30-day volume and talent shortages.

Project Technical Objectives

Core Engineering Goals Accomplished in ElevAI

1. Multi-Format Ingestion

Parse PDF/DOCX resumes server-side up to 5MB, sanitize text buffers, and store securely in private cloud storage.

2. AI ATS & Content Engine

Deploy Google Gemini 2.0 to generate multi-dimensional scoring (0–100) and Google X-Y-Z bullet rewrites.

3. Job Compatibility Engine

Compute real-time role matching scores against live job postings with missing keyword identification.

4. Interactive Mock Interview

Build dynamic Q&A interview engine evaluating technical depth, clarity, and reference responses in real time.

5. Full Recruitment Pipeline

Deliver admin job lifecycle (Draft/Publish/Close), candidate review dossiers, and 6-stage status history audit.

6. Zero-Downtime AI Resilience

Implement local heuristic fallback engines ensuring 100% platform availability during external AI quota spikes.

System at a Glance

Decoupled Three-Tier Architecture & Data Flow

1. Presentation Tier

&›þ React 18 Single Page App

Ø<ß¨ Tailwind CSS UI (Dark Theme)

Ø=Þãþ React Router DOM v7 (SPA)

Ø=ÜÊ Recharts Analytics Visuals

Ø=Ý Axios HTTP Client with JWT Request & Response Interceptors

Ø=Þáþ Client Route Guards (ProtectedRoute & AdminRoute)

2. Application & API Tier

Ø=Þ€ Node.js & Express.js REST API

Ø=Þáþ Middleware Pipeline (JWT Auth, requireAdmin RBAC, RFC 4122 UUID Validation)

&™þ Business Logic Services (Jobs, Resumes, Applications, Analytics)

'(Resilient Dual-Layer AI Engine (Gemini 2.0 + Heuristic Fallback)

Ø=Ý Error Sanitization & Security

3. Data & Cloud Services

Ø=Ü Supabase PostgreSQL (12 Relational Tables & Enums)

&¡ Prisma ORM v6 Client & Type-Safe Query Engine

Ø=ÝÄþ Supabase Storage (Private 'resumes' Bucket & 1h Signed URLs)

Ø>Ý Google Gemini API (gemini-2.0-flash LLM Model)

Technology Stack Matrix

Production-Grade Open Source & Cloud Technologies

Frontend Layer

- React.js 18 (SPA Architecture)
- Tailwind CSS v4 (Glassmorphic Theme)
- Vite 6 (High-Speed Bundler)
- React Router DOM v7
- Recharts (Telemetry Visuals)
- Lucide React Icons

Backend & Server

- Node.js v20+ / v25 Runtime
- Express.js v4.21 REST Server
- Prisma ORM v6 Data Modeling
- Multer (Multipart Memory Uploads)
- PDF-Parse & Mammoth (Extraction)
- Bcrypt.js & JWT Security

Database & Storage

- Supabase PostgreSQL 15+
- 12 Relational Tables & Cascades
- Relational Enums & Unique Keys
- Supabase Cloud Object Storage
- Time-Limited Signed Downloads
- Transaction & Session Pooling

AI & Intelligence

- Google Gemini API (@google/genai)
- LLM Model: gemini-2.0-flash
- Structured JSON Prompt Contracts
- Local Heuristic Offline Engine
- Zero-Downtime Rate Handling
- Parameterized Prompt Security

Security & Auth

- Stateless Signed JWT (7-day)
- 10-Round Salted Bcrypt Hashing
- Server-Side requireAdmin RBAC
- RFC 4122 UUID Parameter Guards
- IDOR Compound Scoping Filters
- Strict Production CORS Limits

Deployment & CI/CD

- Render Web Service (Backend)
- Render Static Site (Frontend)
- SPA Redirect Rewrite (/* /index.html)
- Render YAML Blueprint Config
- Parameterized .env Isolation
- Zero Secrets in Client Bundles

Candidate End-to-End Workflow

Seamless Progression from Onboarding to Application

Step 1: Register

Create account with bcrypt security; establish target Career Goal & experience level.

Step 2: Upload / Build

Upload PDF/DOCX or use Interactive Builder; text is parsed and stored in private vault.

Step 3: ATS Analysis

Receive instant 0–100 ATS Score, section ratings, skill categorization & weaknesses.

Step 4: X-Y-Z Rewrite

Review bullet rewrites formatted with quantifiable impact metrics while preserving facts.

Step 5: Skill Gap

Identify critical missing technologies and follow prioritized learning roadmaps.

Step 6: Mock Interview

Practice role-specific Q&A rounds with instant grading on depth and clarity.

Step 7: Job Fit

Analyze resume against target job description to get match score & missing keywords.

Step 8: Apply & Track

Submit application with score snapshot; track live status (SUBMITTED -> SELECTED).

AI Processing Pipeline & Fault Tolerance

Dual-Layer Architecture Guaranteeing 100% Availability

Ø>Ý Primary AI Pipeline (Gemini 2.0)

1. User Request: Client submits resume/job payload.

2. Server-Side Context Injection: Express backend combines candidate Career Goal, target role, and extracted text.

3. Structured Prompt Engineering: Strictly enforces JSON response contracts with predefined schema schemas.

4. Gemini 2.0 API Execution: Dispatches request to gemini-2.0-flash model.

5. JSON Schema Validation: Validates score bounds (0–100), section arrays, and action items before DB persistence.

Ø=Þáþ Resilient Offline Heuristic Fallback

• The Problem: Upstream free-tier Google Gemini API accounts enforce rate limits (20 requests/day).

• ElevAI Solution: When HTTP 429 or 503 is intercepted, the service automatically diverts to local heuristic calculation engines.

• How It Works: Scans text for 60+ technical competencies, computes keyword density, and generates structured feedback.

• Zero User Impact: Candidates receive complete, actionable reports without application crashes or 500 errors.

Resume Intelligence & Vault Lifecycle

Multi-Format Processing, Versioning & Deletion Safety

1. Upload Ingestion

Supports PDF, DOCX, DOC, and TXT files up to 5MB. Multer handles in-memory file buffers securely.

2. Text Extraction

pdf-parse and mammoth extract clean text strings, stripping binary metadata and formatting artifacts.

3. Cloud Storage Vault

Files saved under user-scoped paths in private Supabase Storage. Downloads require 1-hour signed URLs.

4. Deletion Safety

Resumes attached to active submitted applications are locked to preserve audit integrity for recruiters.

Ø=ÜÝ Interactive Resume Builder & Direct Vault Export

- Multi-Section Editing: Personal Info, Professional Summary, Categorized Skills, Work Experience, Education, Projects, and Custom Sections.
- Live ATS Preview: Instant visual formatting ensuring clean typography and structural readability.
- Direct Vault Export: Compiles builder documents directly into a READY resume in the user's vault for one-click job applications.

Job Marketplace & Application Workflow

Integrated Requisition, Optimization & Submission Flow

1. Admin Posts Job

Requisition created with title, required skills, preferred tech, salary band, and deadline.

2. Targeted Alert

System notifies candidates whose career goals align with the new job opening.

3. Fit Analysis

Candidate analyzes resume against job description, getting match score & missing keywords.

4. One-Click Apply

Application submitted with permanent snapshot of resume and match score (duplicate 409 protected).

Ø=Þáþ Candidate Experience & Application Protection Rules

- No Automatic Rejection: Candidates with lower match scores are never auto-rejected; recruiters review all applicants.
- Deadline Enforcement: Submissions to expired or CLOSED requisitions are automatically rejected by backend validators.
- Immutable Snapshot: If a candidate later edits their resume, the recruiter still sees the exact version submitted.

Admin Recruitment Control Suite

Centralized Candidate Review, Dossiers & Management

Ø=ÜÈ Overview Telemetry

Real-time counters for total users, published jobs, draft requisitions, pending reviews, and selected offers.

Ø=ÜÝ Job Requisitions

Full CRUD suite: create draft jobs, publish requisitions, set salary ranges, and close expired listings.

Ø=Üe Review Dossier

Inspect candidate profile, submitted cover letter, AI match score snapshot, and generate secure resume links.

Ø=Ý Private Notes

Record internal evaluation remarks attached to candidate applications (strictly hidden from user API).

Ø=Ý Pipeline Manager

Transition candidates through recruitment stages (SUBMITTED -> SELECTED) with automated audit logging.

Ø=ÜÊ Analytics Engine

Live PostgreSQL engine: 30-day continuous application trends, skill demand ranking, and deadline radars.

Application Status Lifecycle & Audit Trail

6-Stage Candidate Progression with Immutable History

SUBMITTED

Application received with resume snapshot.

UNDER_REVIEW

Recruiter opens candidate dossier.

SHORTLISTED

Candidate meets key technical criteria.

INTERVIEW

Candidate scheduled for technical round.

SELECTED

Offer extended to successful candidate.

REJECTED

Application archived with decision notice.

Ø=ÜË Immutable Status History Audit Log (`ApplicationStatusHistory`)

- Full Traceability: Every status change creates an immutable event recording oldStatus, newStatus, changedById, and timestamp.
- Candidate Notification Dispatch: Status changes trigger instant notification alerts routed to the applicant's dashboard.
- Multi-Tenant Isolation: Candidates can only view public status transitions; internal recruiter notes are strictly withheld.

Database Entity-Relationship Architecture

12 Relational PostgreSQL Tables Modeled with Prisma ORM

User

- id (UUID, PK)
- email (Unique)
- passwordHash
- role (USER/ADMIN)

CareerGoal

- id (UUID, PK)
- userId (FK, Unique)
- targetRole
- experienceLevel

Resume

- id (UUID, PK)
- userId (FK)
- storagePath
- extractedText
- status (READY)

Job

- id (UUID, PK)
- createdBy (FK)
- title, description
- requiredSkills
- status (PUBLISHED)

Application

- id (UUID, PK)
- jobId (FK)
- userId (FK)
- resumeId (FK)
- matchScore, status

StatusHistory

- id (UUID, PK)
- applicationId (FK)
- oldStatus, newStatus
- changedById (FK)

ApplicationNote

- id (UUID, PK)
- applicationId (FK)
- adminId (FK)
- note (Text)

AI Records

- ResumeAnalysis
- ResumeImprovement
- JobOptimization
- SkillGapAnalysis

Backend Request & API Execution Lifecycle

Layered Defense, Validation & Processing Pipeline

1. Inbound

Browser dispatches HTTPS request with Bearer JWT header.

2. Headers

Express applies X-Frame-Options: DENY, nosniff, and CORS.

3. Auth Mid

Verifies cryptographically signed JWT; attaches req.user.

4. UUID Check

validateUUIDParams checks RFC 4122 regex; rejects SQLi.

5. Services

Executes Prisma queries or Gemini AI processing.

6. JSON Out

Returns formatted response; masks database errors.

Ø=Þáþ Real-World Trace: Candidate Submits Job Application (`POST /api/applications`)

1. JWT Authenticated -> 2. Verify jobId & resumeId UUIDs -> 3. Ensure Job is PUBLISHED & deadline active -> 4. Confirm Resume ownership -> 5. Check duplicate application (409 if exists) -> 6. Store Application with matchScore -> 7. Append SUBMITTED to status history -> 8. Return HTTP 201 Created.

Multi-Layered Security & RBAC Enforcement

Comprehensive Protections Shielding Application Data

Ø=Ý Password Security

10-round salted bcrypt hashing ensures raw passwords are never stored.

Ø=Þáþ Server-Side RBAC

requireAdmin middleware validates JWT role, blocking unauthorized candidates.

Ø=Þ« IDOR Elimination

Every database query filters by authenticated userId, preventing cross-tenant access.

Ø=Þáþ Parameter Hardening

RFC 4122 regex validation on all route IDs blocks SQL injection & path traversal.

Ø=ÝÄþ Private Storage

Supabase Storage bucket is private; downloads require 1-hour signed URLs.

Ø=Ý Zero Client Secrets

Backend API keys (Gemini, Supabase Service Role) are never exposed to browser.

Smart Notifications & Real Analytics Engine

Live Database Telemetry Without Synthetic Numbers

Ø=Ý Smart Notification System

• Event-Driven Routing: Instant alerts dispatched when matching jobs are published or application statuses update.

• Live Bell Badge Counter: Unread indicator automatically updates and decrements upon mark-as-read.

• Multi-Tenant Isolation: User B is strictly blocked from reading, modifying, or deleting User A notifications.

• User Preference Controls: Candidates can toggle notification categories (Jobs, Resumes, Reminders).

Ø=ÜÊ Recruitment Analytics Engine

• 100% Real Database Data: All metrics calculated from live PostgreSQL records without hardcoded numbers.

• 30-Day Continuous Trends: Daily time-series buckets ensure empty days reflect 0 submissions rather than gaps.

• Recruitment Funnel: Real progression rates from SUBMITTED -> UNDER_REVIEW -> SHORTLISTED -> SELECTED.

• Skill Demand Radar: Aggregates top required skills across all active postings to identify talent trends.

UI/UX Design & Product Experience

Cyber Glassmorphism with Fluid Responsiveness

Ø<ß Glassmorphism

Dark slate palette (#0A0E17) with translucent cards, 1px subtle borders, and soft backdrop blurs.

Ø=Üñ Fully Responsive

Adaptive layouts across mobile (320px), tablet, laptop, and ultra-wide 4K desktop screens.

&¡ Micro-Animations

Smooth Recharts visual progressions, hover glow transitions, and responsive modal dialogues.

Ø=Ý Contextual Toasts

Interactive feedback system with success, error, warning, and loading status states.

Ø<ß›þ Tabbed Views

Clean separation between Candidate tools, Resume vaults, Job feeds, and Admin control suites.

&• Accessible Controls

Semantic HTML5 structure, ARIA accessibility attributes, and high-contrast readable typography.

Testing & Master Verification Status

100% Pass Rate Across Automated Full-Lifecycle Audits

Auth & Credentials

Result: PASS (8/8)

Admin RBAC Checks

Result: PASS (3/3)

Resume Vault & Locks

Result: PASS (7/7)

AI Fallback Resilience

Result: PASS (4/4)

Job Lifecycle Rules

Result: PASS (6/6)

Application Tracking

Result: PASS (6/6)

Admin Review Dossier

Result: PASS (9/9)

Analytics Calculations

Result: PASS (5/5)

Parameter Hardening

Result: PASS (5/5)

Client Vite Build

Result: PASS (14s)

API Health Endpoints

Result: PASS (200)

OVERALL VERIFICATION

Result: 54/54 PASS

Current Limitations & Future Roadmap

Operational Realities & Strategic Product Roadmap

& þ Current System Limitations

1. Upstream AI Rate Quotas: Google Gemini free-tier accounts enforce a 20 request/day quota. (Mitigated by ElevAI's built-in offline heuristic fallback engine).

2. Browser Web Speech Dependency: Voice input during mock interviews depends on browser Web Speech API support (Chrome/ Edge recommended).

3. Single Requisition Attachments: Applications currently link one primary resume version per job submission.

Ø=Þ€ Future Scope & Strategic Roadmap

• Out-of-Band Email Alerts: Integration with SendGrid / AWS SES for automated email notifications on status updates.

• Embedded WebRTC Video Interviews: Direct browser-to-browser video conferencing within the Admin Review Suite.

• Bulk Recruiter Resume Parsing: Multi-resume batch uploads and automated talent matching for high-volume hiring.

• Native Mobile Apps: Cross-platform React Native applications for iOS and Android devices.

ElevAI — Final Summary

A Production-Ready Career Intelligence & Recruitment Platform

Ø<ß¯ Key Engineering Achievements

' Complete Candidate Career Suite: Multi-format ATS analysis, X-Y-Z content rewrites, job fit optimization, and interactive mock interviews.

' Full Recruiter Control: Job requisition management, candidate review dossiers, private recruiter notes, and real PostgreSQL analytics.

' Robust Decoupled Architecture: React 18 SPA + Node/Express REST API + Supabase PostgreSQL + Google Gemini 2.0 AI.

' Zero-Downtime Resilience: Built-in heuristic fallback engine ensuring 100% platform availability during AI quota spikes.

' Multi-Layered Security: Strict server-side RBAC, IDOR compound filters, parameter hardening, and private cloud storage signed URLs.

ElevAI — "Your Career, Elevated."